WASP-43b

R-0634

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_170428` · created 2026-07-25T17:13:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457871.672 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.86 +/- 2.23 [%]
Orbital Inclination (inc)	80.52 +/- 2.82
Airmass coefficient 1 (a1)	5.97 +/- 3.6
Airmass coefficient 2 (a2)	-0.94 +/- 0.48
Scatter in the residuals of the lightcurve fit is	55.13 %
Best Comparison Star	None
Optimal Aperture	1.87
Optimal Annulus	8.01
Transit Duration (day)	0.044 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.057 vs 0.1594 (deviation 0.45σ)
Duration fit vs published	0.044 d vs 0.0512 d (deviation 0.34σ)
Mid-time O–C	-2.9 min
Depth SNR	2.4
Residual scatter	55.13 %

Light curve

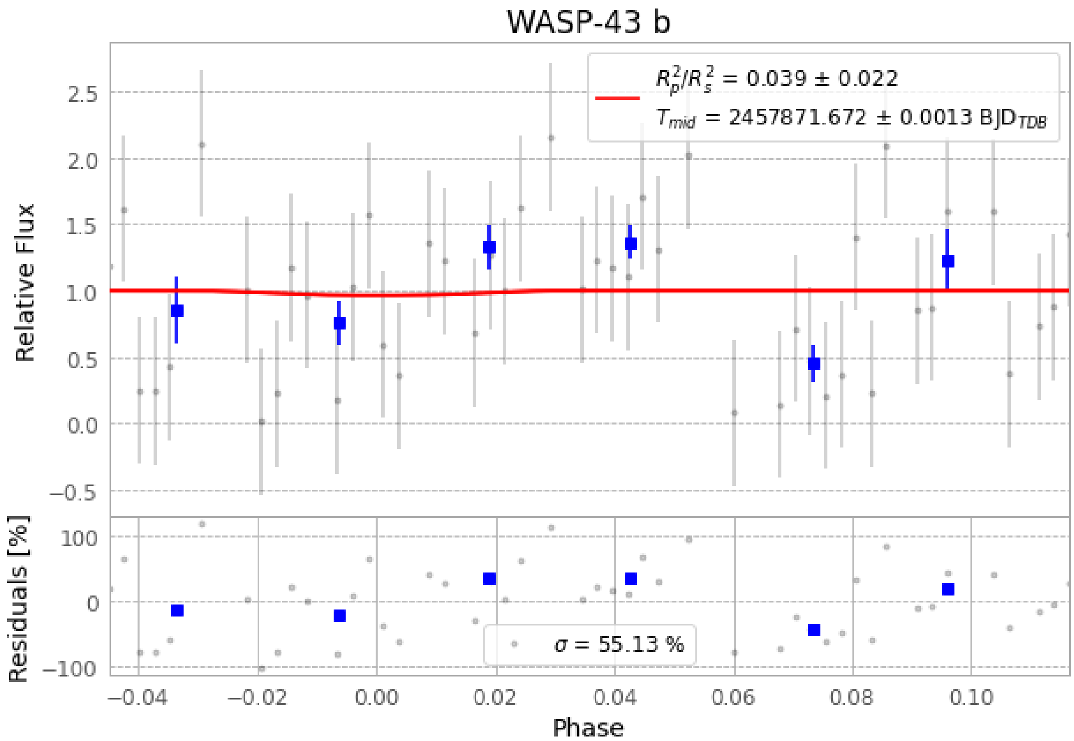

Figure 1 — FinalLightCurve_WASP-43 b_2017-04-27.png (SHA-256 9fec721256b0cf9f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 431], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cd03897d38716a2d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

66 FITS frames (43 MB), `WASP-43170428030912.FITS` → `WASP-43170428062412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170428.log (7e54370c196a…), FinalLightCurve_WASP-43 b_2017-04-27.png (9fec721256b0…), FinalLightCurve_WASP-43 b_2017-04-27.csv (b258b9bd2098…)

Compute resources

Wall time 160.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)